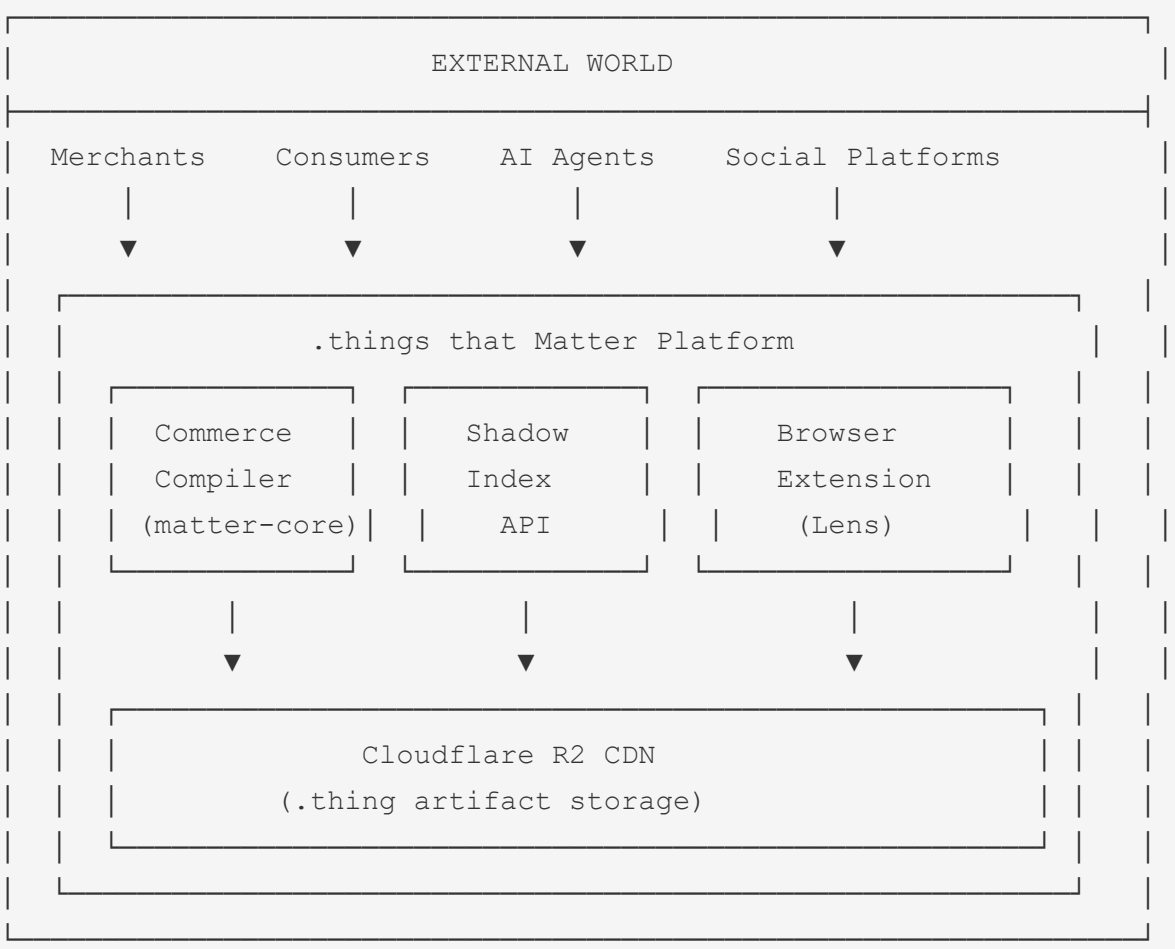
.things that Matter

Technical Architecture Document

Document Information	
---	---
Company Name	**.things that Matter**
Document Type	**Technical Architecture**
Version	**1.0**
Date	**January 21, 2026**
Author	**Matome Mboweni, CTO**

1. Architecture Overview

1.1 System Context



1.2 Core Design Principles

Principle	Description
Decentralization	**Products are self-sovereign; no central truth server required**
Efficiency	**Binary format optimized for machine parsing (O(1) field access)**
Portability	**Artifacts travel across any medium (images, links, files)**

Cryptographic Trust	**Ed25519 signatures replace platform trust**

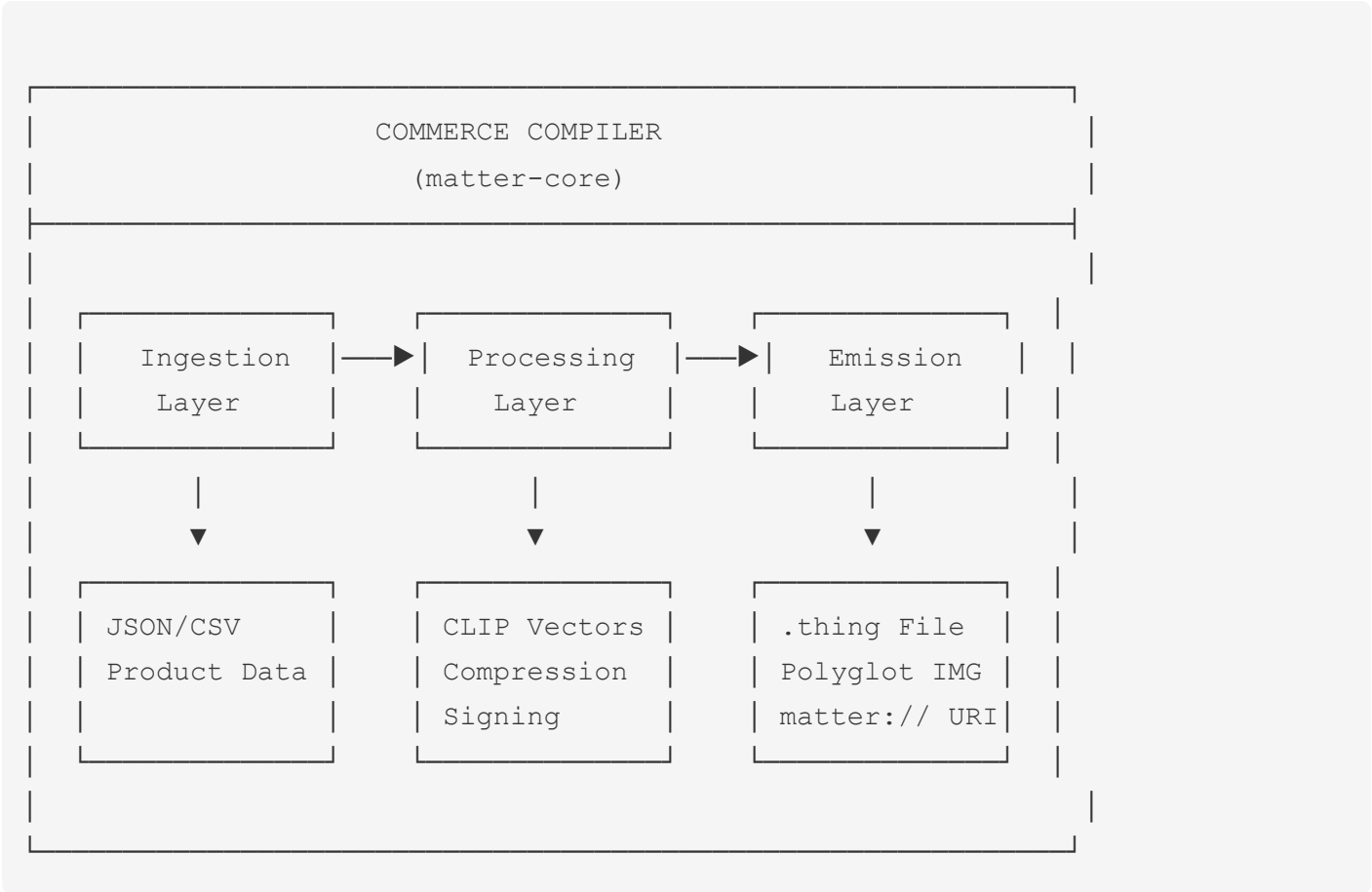
Human + Agent Alloy	**Dual optimization for visual appeal and computational efficiency**

---

2. Component Architecture

2.1 Commerce Compiler (matter-core)

The heart of the system. Transforms product data into Universal Commerce Containers.



Technology Stack: - **Language:** Rust (Python MVP first, then port) - **Serialization:** Google FlatBuffers - **Compression:** Zstandard (Zstd) - **Cryptography:** Ed25519 (via `ed25519-dalek` crate) - **WASM:** Compiled for browser/Node.js via `wasm-pack`

Key Operations:

Operation	Description	Output
`compile()`	Full pipeline: ingest → process → emit	.thing + image + URI
`sign()`	Apply Ed25519 signature	Signed .thing
`verify()`	Validate signature integrity	Boolean
`vectorize()`	Generate CLIP embedding	512-dim float vector
`embed()`	Steganographic image embedding	Polyglot image

2.2 FlatBuffers Schema

Binary format for Universal Commerce Containers.

```
flatbuffers
// thing.fbs - Universal Commerce Container Schema

namespace Matter;

table Product {
  // Identity
  id: string (required);
  hash: [ubyte] (required);        // SHA-256 of canonical form
  version: uint8 = 1;

  // Commerce Data
  title: string (required);
  description: string;
  price: uint64;                   // Price in cents (avoid floats)
  currency: string = "USD";
  sku: string;

  // Media
  image_url: string;
  thumbnail: [ubyte];              // Embedded thumbnail (optional)

  // Semantic Layer (Cognitive)
  clip_vector: [float];            // 512-dim CLIP embedding
  categories: [string];
  tags: [string];

  // Commerce Logic
  checkout_url: string (required);
  merchant_id: string;
  inventory_hint: InventoryStatus = UNKNOWN;

  // Metadata
  created_at: uint64;              // Unix timestamp
  expires_at: uint64;              // TTL for freshness
  source_url: string;

  // Cryptographic Seal
  signature: [ubyte];              // Ed25519 signature
  signer_pubkey: [ubyte];          // Public key for verification
}
```

```
enum InventoryStatus : byte {
  UNKNOWN = 0,
  IN_STOCK = 1,
  LOW_STOCK = 2,
  OUTOFSTOCK = 3
}

table ThingFile {
  magic: uint32 = 0x54484E47;      // "THNG" magic bytes
  version: uint8 = 1;
  product: Product (required);
  gist: Gist;                      // Minimal data for quick reads
}

table Gist {
  title: string;
  price: uint64;
  currency: string;
  image_url: string;
}

root_type ThingFile;
```

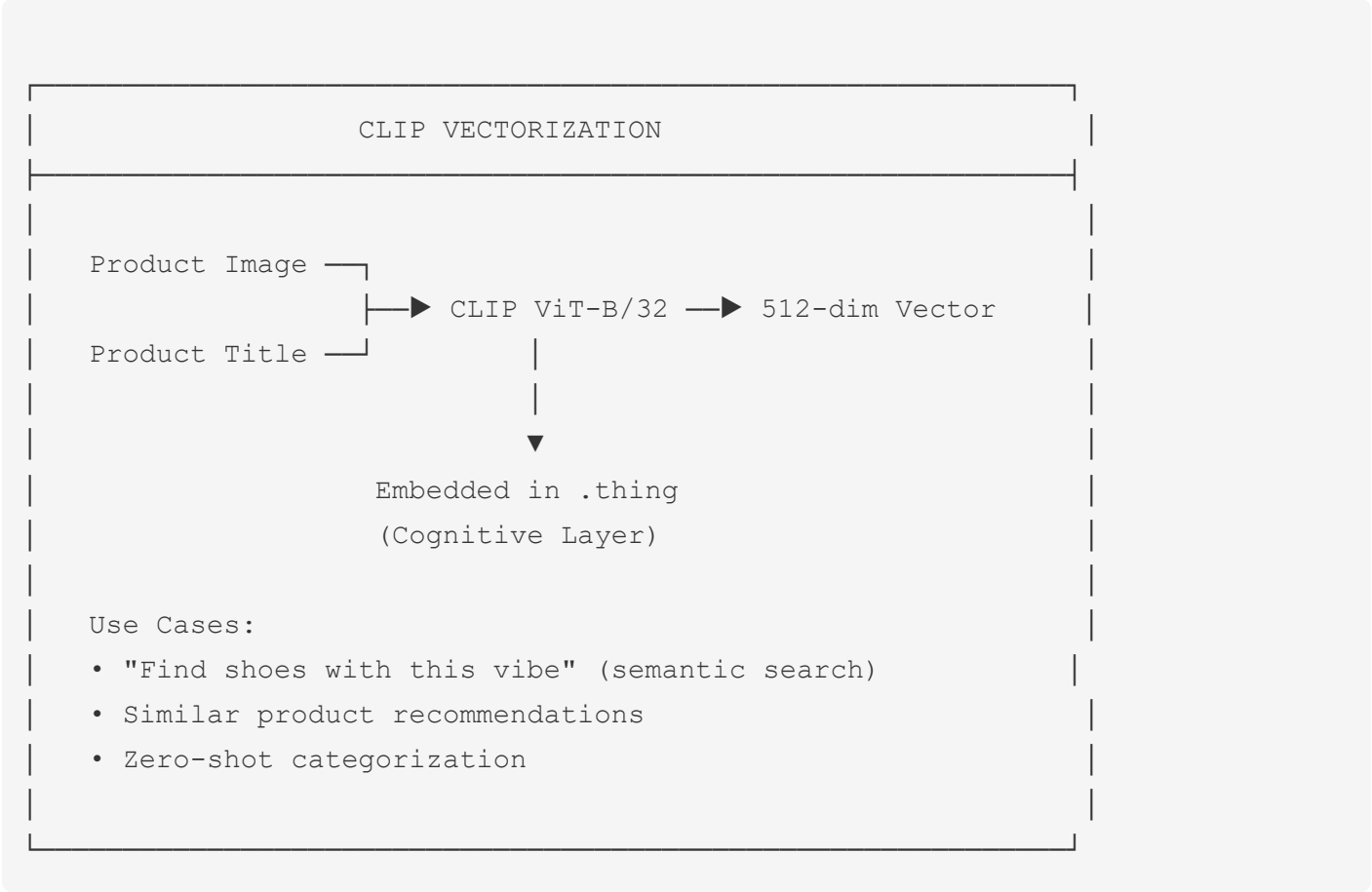
Why FlatBuffers over JSON/Protobuf?

Feature	JSON	Protobuf	FlatBuffers
Parse time	**O(n)**	**O(n)**	**O(1)**
Memory allocation	**High**	**Medium**	**Zero-copy**
Random field access	**No**	**No**	**Yes**
Human readable	**Yes**	**No**	**No**

Schema evolution	**Poor**	**Good**	**Good**

Zero-Copy Access: AI agents can read `price` directly from byte offset without parsing entire file.

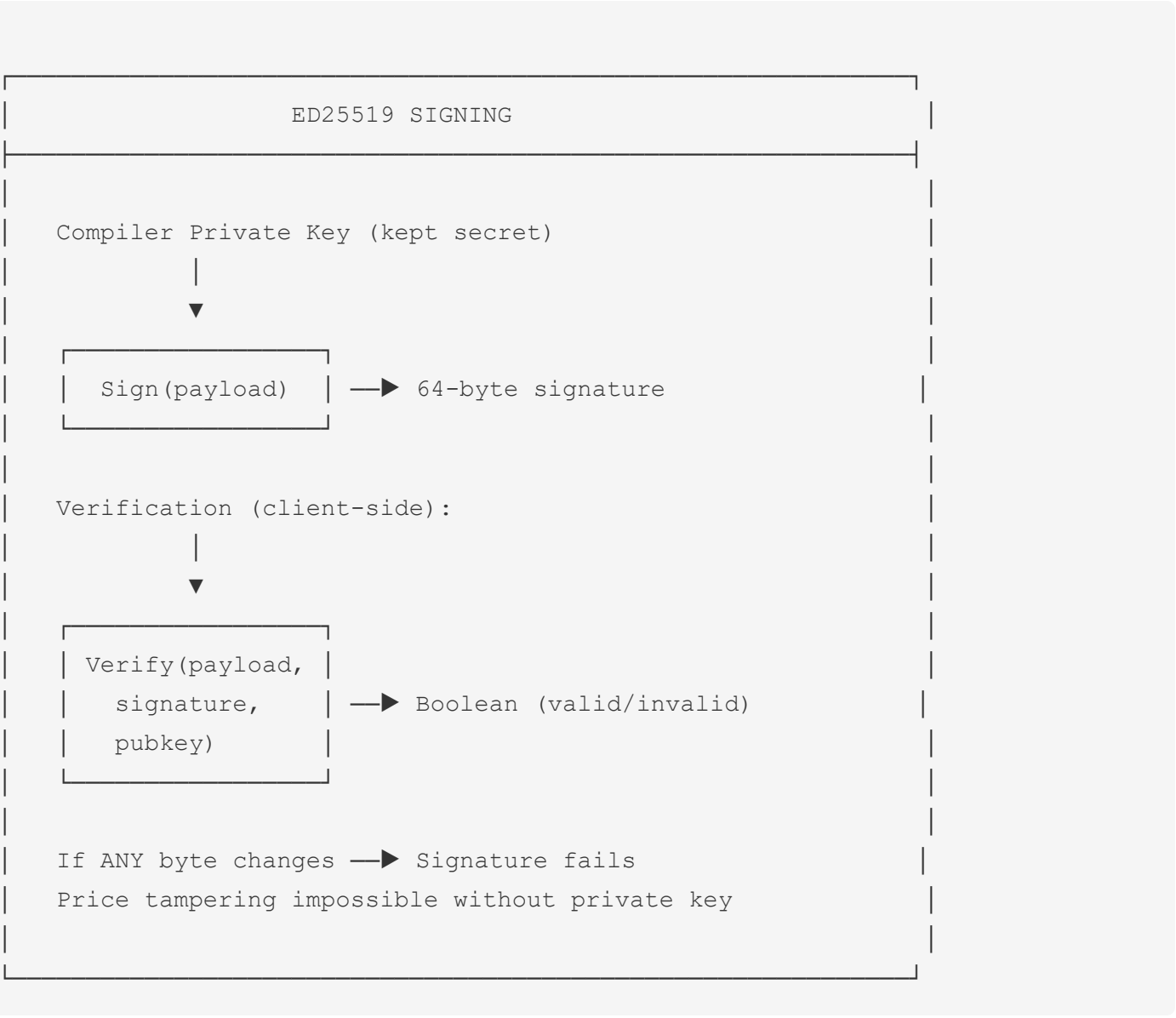
2.3 Semantic Vectorization (CLIP Integration)



Implementation: - **Service:** Replicate API (serverless CLIP) - **Model:** ViT-B/32 (OpenAI CLIP) - **Cost:** ~R0.40 per image - **Latency:** ~200ms per inference

Vector Storage: - Embedded directly in .thing file header - Enables local similarity search without API calls - 512 floats × 4 bytes = 2KB per product

2.4 Cryptographic Signing

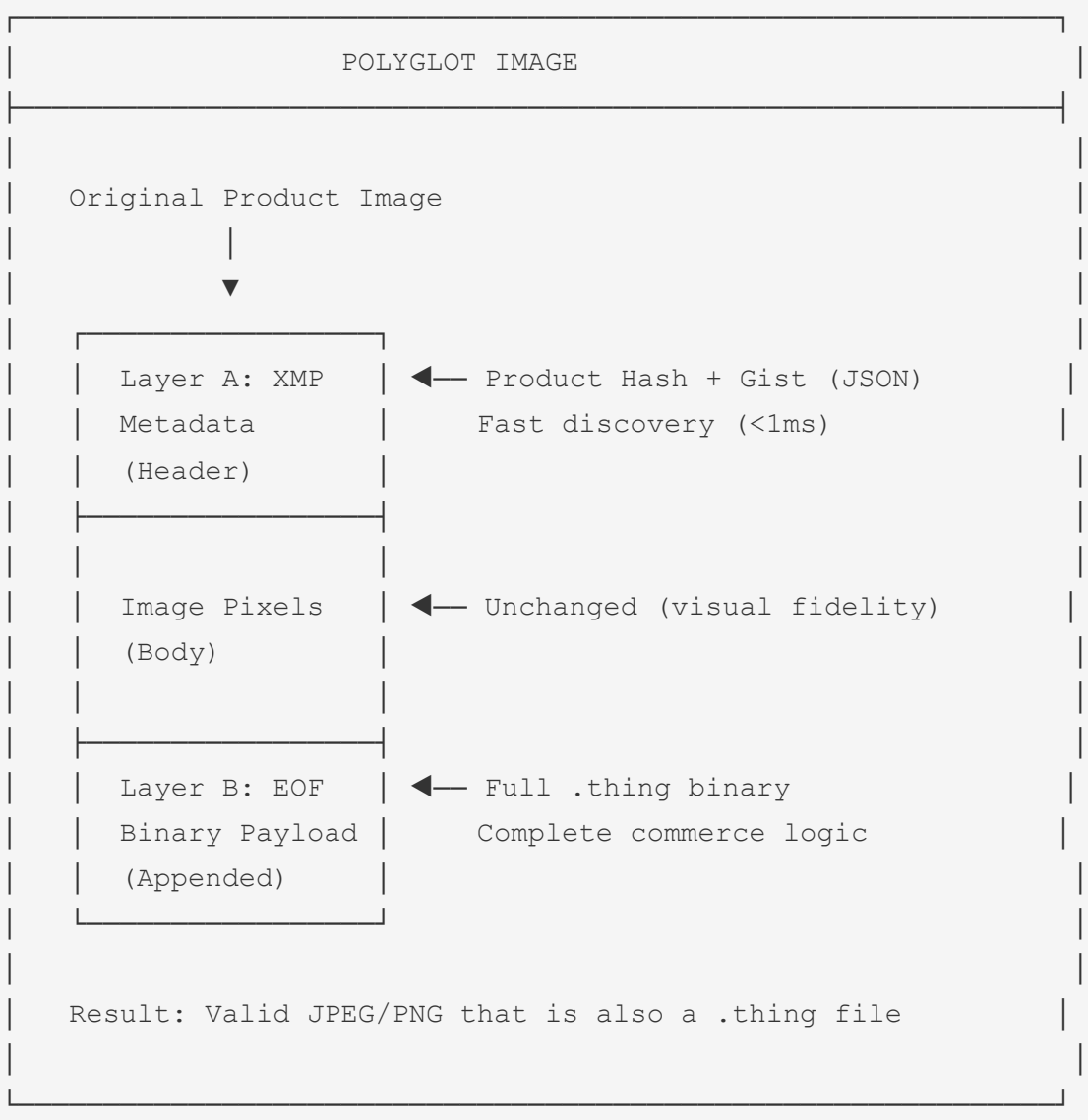


Why Ed25519? - Fast verification (important for browser extension) - Small signature size (64 bytes) - Deterministic signatures - Well-audited, no ECDSA pitfalls

Key Management: - Private key stored in secure environment variable - Public key distributed with extension/SDK - Key rotation: new key = new signature version

2.5 Distribution Matrix

2.5.1 Polyglot Image (Steganography)



XMP Metadata Beacon (Layer A):

```
xml



      a1b2c3d4...
      {"title":"Product","price":2999,"currency":"USD"}
      1
```

Platform Survival Strategy:

Platform	XMP Survives	EOF Survives	Strategy
---	---	---	---
Discord	Yes	Yes	Full polyglot works
Telegram	Yes	Yes	Full polyglot works
Twitter/X	No	No	Use matter:// URI
Instagram	No	No	Use matter:// URI
WhatsApp	Partial	No	XMP gist + matter://
Email	Yes	Yes	Full polyglot works
Direct download	Yes	Yes	Full polyglot works

2.5.2 Matter URI Protocol

```
matter://a1b2c3d4e5f6...

Resolution Flow:
1. Extension intercepts matter:// link
2. Extract hash from URI
3. Query Shadow Index API: GET /resolve/{hash}
4. Receive CDN URL for .thing file
5. Fetch and verify .thing
6. Display product overlay
```

URI Format:

```
matter://{base58_hash}[?params]

Examples:
matter://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG
matter://QmYwAPJzv5CZsnA625s3Xf2nemtYgPpHdWEz79ojWnPbdG?ref=twitter
```

2.5.3 Raw .thing File

Direct file distribution for: - Developer integrations - Bulk data exchange - Offline catalogs - AI agent consumption

MIME Type: `application/vnd.matter.thing` **File Extension:** `.thing`

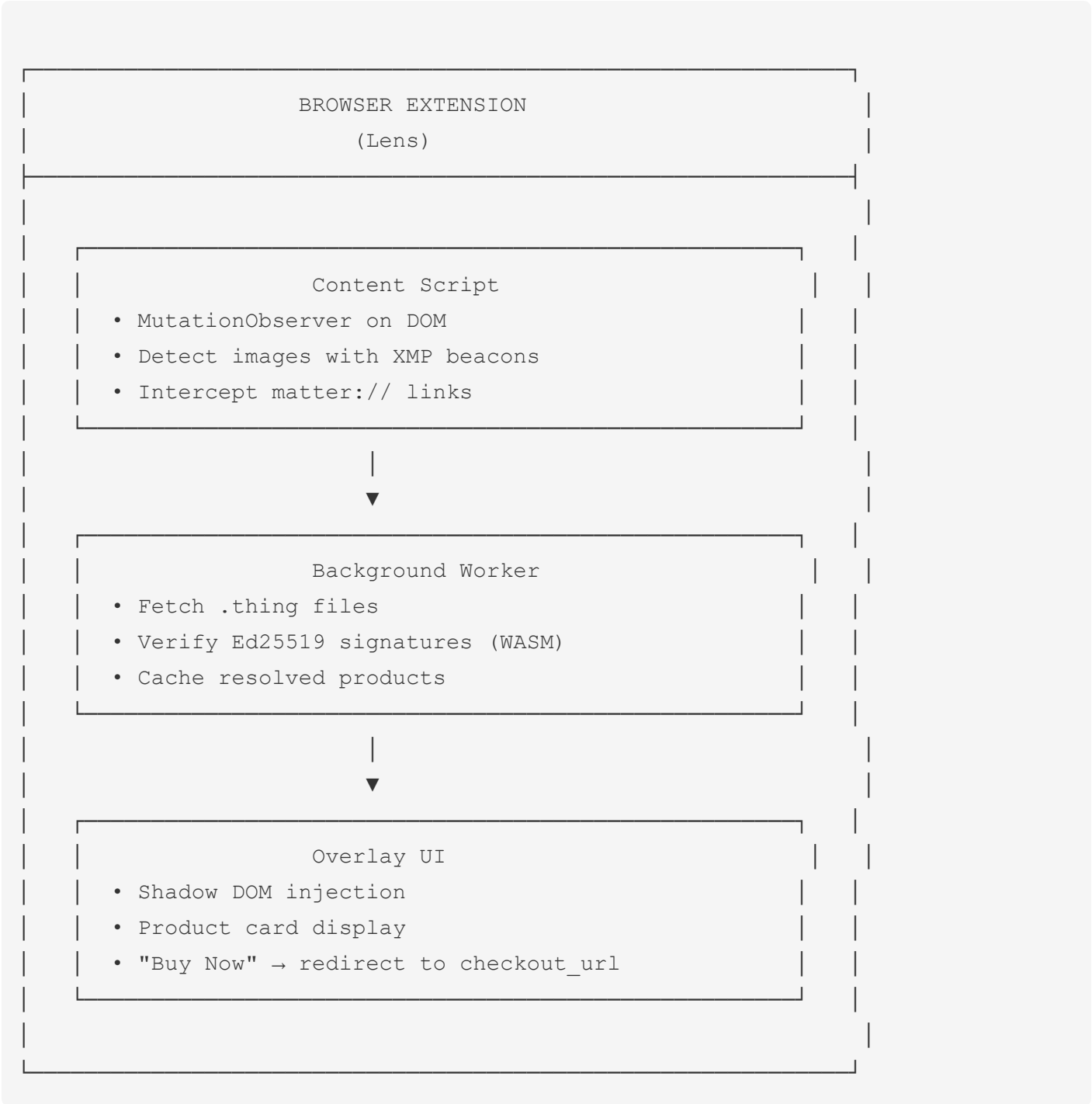
2.6 Shadow Index API

Fast hash-to-artifact resolution service.

```
┌─────────────────────────────────────────────────────────────┐
│                    SHADOW INDEX API                         │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│   Endpoints:                                                │
│                                                             │
│   POST /compile                                             │
│   Body: { product_json }                                    │
│   Response: { hash, thingurl, imageurl, matter_uri }        │
│                                                             │
│   GET /resolve/{hash}                                       │
│   Response: { thingurl, gist, expiresat }                   │
│                                                             │
│   GET /gist/{hash}                                          │
│   Response: { title, price, currency, image_url }           │
│                                                             │
│   POST /batch                                               │
│   Body: { products: [...] }                                 │
│   Response: { results: [...] }                              │
│                                                             │
└─────────────────────────────────────────────────────────────┘
```

Infrastructure: - **Hosting:** Vercel Edge Functions (free tier) - **Database:** Supabase (hash → metadata mapping) - **Storage:** Cloudflare R2 (artifact files) - **Cache:** Redis Cloud free tier (hot lookups)

2.7 Browser Extension (Lens)



Technology Stack: - **Framework:** Vanilla JS (minimize bundle size) - **Crypto:** libmatter compiled to WASM - **UI:** Shadow DOM (isolated styles) - **Storage:** IndexedDB (local cache)

---

3. Data Flow

3.1 Compilation Flow (Merchant → Artifact)

```
1. Merchant submits product JSON
         |
         ▼
2. Compiler validates input schema
         |
         ▼
3. Fetch product image, send to CLIP API
         |
         ▼
4. Receive 512-dim semantic vector
         |
         ▼
5. Compress description with Zstd
         |
         ▼
6. Serialize to FlatBuffers binary
         |
         ▼
7. Sign with Ed25519 private key
         |
         ▼
8. Generate polyglot image (XMP + EOF)
         |
         ▼
9. Upload artifacts to Cloudflare R2
         |
         ▼
10. Index hash in Shadow Index DB
         |
         ▼
11. Return: { hash, thing_url, image_url, matter_uri }
```

3.2 Discovery Flow (Consumer finds product)

```
1. Consumer sees image on social media
        │
        ▼
2. Lens extension detects XMP beacon
        │
        ▼
3. Extract hash from beacon
        │
        ▼
4. Check local cache for .thing
        │
        ├── Cache hit: Use cached data
        │
        └── Cache miss: Fetch from CDN
                │
                ▼
5. Verify Ed25519 signature
        │
        ├── Invalid: Show warning, block
        │
        └── Valid: Continue
                │
                ▼
6. Render product overlay (Shadow DOM)
        │
        ▼
7. User clicks "Buy Now"
        │
        ▼
8. Redirect to merchant checkout_url
```

---

4. Security Architecture

4.1 Threat Model

Threat	Attack Vector	Mitigation
Price tampering	Modify .thing file	Ed25519 signature invalidates
Fake products	Create malicious .thing	Only signed by our key accepted
Man-in-middle	Intercept CDN response	HTTPS + signature verification
XSS via overlay	Inject malicious HTML	Shadow DOM isolation, CSP
Key compromise	Steal signing key	Key rotation, monitoring

4.2 Trust Model

```
Trust Anchor: .things that Matter signing key
         |
         ▼
Verified: Compiler signs all .thing files
         |
         ▼
Client: Extension only trusts our public key
         |
         ▼
Result: Tamper-proof commerce data
```

4.3 Security Controls

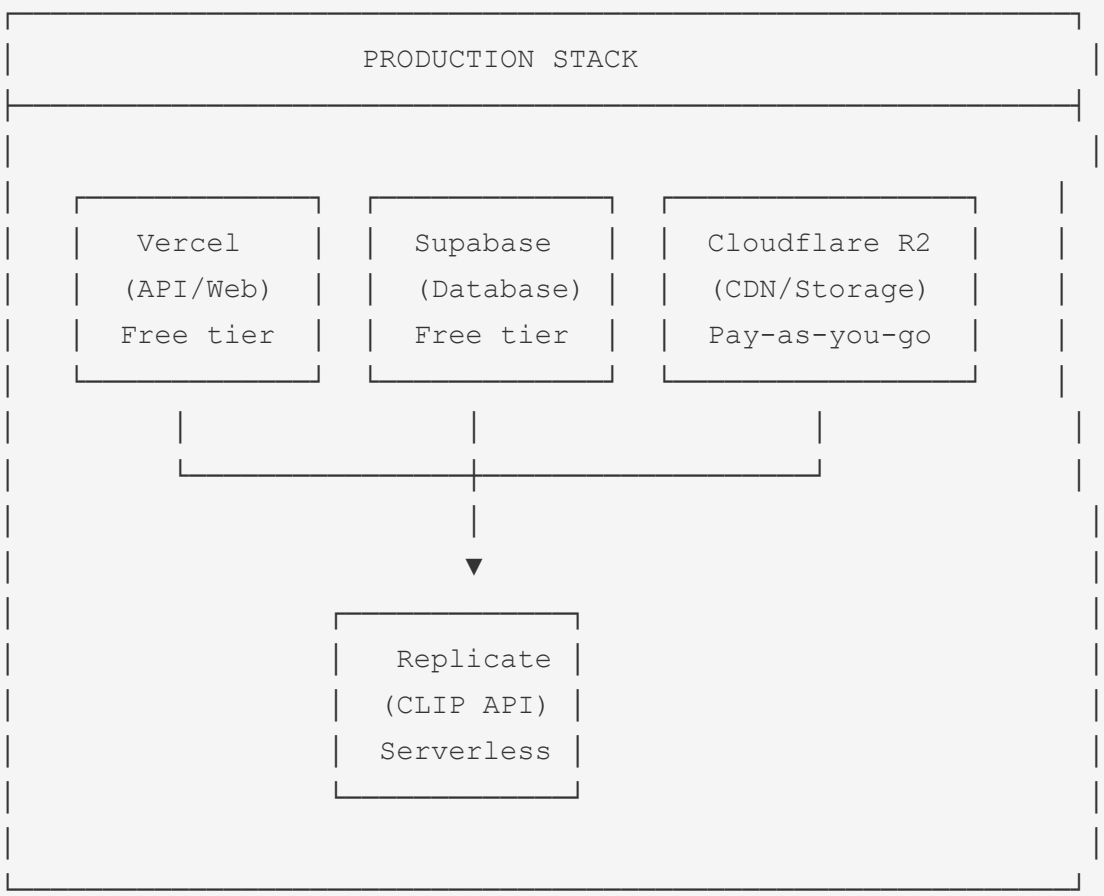
Control	Implementation

| --- | --- |
Input validation	**Zod schema validation**
Output encoding	**Strict FlatBuffers schema**
Cryptographic signing	**Ed25519**
Transport security	**HTTPS everywhere**
DOM isolation	**Shadow DOM for overlays**
Content Security Policy	**Strict CSP in extension**

---

5. Infrastructure

5.1 Production Architecture



5.2 Service Dependencies

Service	Purpose	Tier	Monthly Cost
---	---	---	---
Vercel	**API hosting, web app**	**Hobby (Free)**	**R0**
Supabase	**Database, auth**	**Free**	**R0**
Cloudflare R2	**Artifact storage**	**Pay-as-you-go**	**~R630/mo**
Replicate	**CLIP inference**	**Pay-per-use**	**~R540/mo**

Redis Cloud	**Caching**	**Free**	**R0**

Sentry	**Error tracking**	**Free**	**R0**

---

6. Development Standards

6.1 Code Organization

```
things-that-matter/
├── packages/
│   ├── matter-core/          # Rust compiler library
│   │   ├── src/
│   │   │   ├── lib.rs
│   │   │   ├── schema/       # FlatBuffers definitions
│   │   │   ├── compiler/     # Compilation pipeline
│   │   │   ├── crypto/       # Ed25519 signing
│   │   │   ├── steg/         # Steganography engine
│   │   │   └── clip/         # CLIP integration
│   │   ├── Cargo.toml
│   │   └── README.md
│   │
│   ├── matter-py/            # Python bindings (MVP)
│   │   ├── matter/
│   │   ├── pyproject.toml
│   │   └── README.md
│   │
│   ├── shadow-index/         # API service
│   │   ├── api/
│   │   ├── package.json
│   │   └── README.md
│   │
│   └── lens-extension/       # Browser extension
│       ├── src/
│       ├── manifest.json
│       └── README.md
│
├── docs/                     # Documentation
├── scripts/                  # Build/deploy scripts
└── README.md
```

6.2 Technology Choices

Component	Technology	Rationale

Compiler	**Rust**	**Memory safety, WASM compilation**
MVP	**Python**	**Fast iteration, familiar**
API	**TypeScript/Node**	**Vercel compatibility**
Extension	**Vanilla JS**	**Minimal bundle size**
Schema	**FlatBuffers**	**Zero-copy access**
Storage	**Cloudflare R2**	**S3-compatible, cheap egress**
Database	**PostgreSQL (Supabase)**	**Reliable, free tier**

6.3 Quality Standards

Metric	**Target**
---	---
Test coverage	**>70% for compiler**
Documentation	**All public APIs documented**
Code review	**Self-review + open-source community**
Performance	**<500ms compile time**
Bundle size	**<100KB extension**

7. Appendices

7.1 API Reference (Shadow Index)

See separate API documentation.

7.2 FlatBuffers Schema

See `schema/thing.fbs` in repository.

7.3 Extension Manifest

See `lens-extension/manifest.json` in repository.

Document Version: 1.0 **Last Updated:** January 21, 2026 **Next Review:** Post-Phase 0 (March 2026)